EXCEL FOR DATA ANALYTICS · TELUGU YOUTUBE TUTORIAL

E-Commerce Data Cleaning, Validation & Dashboard Project

End-to-End · Beginner Friendly · Industry Relevant

🔍 Raw Data | 🧹 Data Cleaning | ✅ Validation | 📊 Charts | 📈 Dashboard | 🎯 Assignment

500	22	5	1
Raw Data Rows	Columns	Cleaning Steps	Dashboard

What We'll Cover Today — Agenda

AGENDA

Oka video lo complete end-to-end Excel analytics project. 500-row messy e-commerce data nundi professional dashboard varaku step-by-step nerchukuntam!

01 Dataset Overview

500-row Flipkart-style messy data — 22 columns, all issues listed

02 Data Cleaning (6 Steps)

TRIM, PROPER, ISBLANK, Date Fix, Category Standardize, Number Validation

03 Data Validation

Dropdowns, number rules, date rules, custom formula validation

04 Data Visualization

5 chart types — Column, Line, Pie, Bar, Donut + Chart Selection Guide

05 Final Dashboard

KPI cards, Pivot Tables, Pivot Charts, Slicers, Timeline — interactive!

06 Assignment

100-mark student project — practice all skills learned today

Why Data Cleaning Matters — Real World Problems

CONTEXT

Real-world data is ALWAYS messy. 80% of a data analyst's time is spent cleaning data. Dirty data = wrong decisions = job risk. Idi mana first skill master cheyyali!

Duplicate Records

Impact: Totals doubled — wrong sales figures

Example: ORD-1001 appears TWICE in dataset

Blank / Null Values

Impact: SUM, VLOOKUP, COUNTIF all break

Example: Customer Name = empty in 15 rows

Inconsistent Casing

Impact: COUNTIF misses matches

Example: 'hyderabad', 'HYDERABAD', 'Hyderabad'

Wrong Date Formats

Impact: Sort & filter completely wrong

Example: 15-Jan-2024, 15/01/24, 2024-01-15

Invalid Categories

Impact: Pivot Table shows wrong groups

Example: 'Electronicss', 'electronic', 'ELECTRONICS'

Negative/Text Numbers

Impact: SUM & AVERAGE give wrong results

Example: Quantity = 'two', Price = '₹35,000'

Dataset Overview — 500-Row E-Commerce Sales Data

DATASET

Flipkart-style Indian e-commerce dataset — 500 orders, 22 columns — intentionally messy with all common real-world data quality issues.

Column	Data Type	Example Value	Issue Present
Order_ID	Text	ORD-1001	20 Duplicates
Order_Date	Date	15-Jan-2024	3 Different Formats
Customer_Name	Text	Ravi Kumar	Extra Spaces, Wrong Case
Region	Text	South	Invalid: 'Soth', 'SOUTH'
Category	Text	Electronics	Typos: 'Electronicss'
Quantity	Number	2	Negative values, 'two'
Unit_Price	Number	35000	Blank cells, '₹35,000'
Discount_%	Number	10	Values >100, text '10%'

DATA QUALITY ISSUES SUMMARY

- **~20** Duplicate Rows
- **~35** Blank Values
- **~80** Wrong Casing
- **~30** Bad Date Formats
- **~25** Invalid Categories
- **~15** Negative Numbers
- **~12** Text in Numbers

SECTION 02

Data Cleaning

6 Steps — TRIM · PROPER · ISBLANK · Date Fix · Categories · Numbers

Step 1 — Remove Duplicates

CLEANING

Duplicate records inflate your totals and distort all analysis. Step 1 is always to identify and remove them before any other cleaning.

SYNTAX / STEPS

```
=IF(COUNTIF($A$3:$A2,A3)>1,"DUPLICATE","UNIQUE")

→ Add as helper column to flag duplicates
```

Data tab → Remove Duplicates → Select Order_ID → OK

```
=COUNTA(A3:A502)-SUMPRODUCT((COUNTIF(A3:A502,A3:A502)>1
)/COUNTIF(A3:A502,A3:A502))

→ Count remaining duplicates after removal
```

EXAMPLE DATA

Order_ID	Customer	Dup_Flag
ORD-1001	Ravi Kumar	UNIQUE
ORD-1002	Priya Sharma	UNIQUE
ORD-1001	Ravi Kumar	DUPLICATE ←
ORD-1003	Amit Singh	UNIQUE
ORD-1002	Priya Sharma	DUPLICATE ←

FORMULA

```
// Flag duplicates first:
=IF(COUNTIF($A$3:$A2,A3)>1,
   "DUPLICATE","UNIQUE")

// Count total dupes:
=SUMPRODUCT(
  (COUNTIF(A3:A502,A3:A502)>1)/
  COUNTIF(A3:A502,A3:A502))
```

✏️ EXERCISE

Q1: Add column W 'Dup_Flag' → write COUNTIF formula

Q2: Filter 'DUPLICATE' rows → count how many

Q3: Use Data → Remove Duplicates → Order_ID only → record removed count

💡 *ALWAYS flag first, review, then delete. Never blindly run Remove Duplicates without checking!*

Step 2 — TRIM & CLEAN — Fix Extra Spaces

CLEANING

Extra spaces break VLOOKUP, COUNTIF, and pivot tables — ' Ravi Kumar' ≠ 'Ravi Kumar'. TRIM removes them all. CLEAN removes hidden characters from imports.

SYNTAX / STEPS

```
=TRIM(text)              → =TRIM("  Sneha Reddy  ") → "Sneha Reddy"
=CLEAN(text)             → removes non-printable characters
=TRIM(CLEAN(C3))         → best practice: combine both
=PROPER(TRIM(CLEAN(C3))) → clean + title-case in one formula
=IF(TRIM(C3)<>C3,"Dirty","Clean") → detect dirty cells
```

FORMULA

```
// Complete cleaning formula:
=PROPER(TRIM(CLEAN(C3)))

// Detect dirty cells:
=LEN(C3)-LEN(TRIM(C3))
→ count of extra spaces
```

EXAMPLE DATA

Original	TRIM Result	LEN Before	LEN After
Sneha Reddy	Sneha Reddy	14	10
RAVI KUMAR	Ravi Kumar	10	10
amit	Amit	8	4
Kabir Mehta	Kabir Mehta	11	11

✏️ EXERCISE

Q1: In Clean_Data!C3: =PROPER(TRIM(CLEAN(Raw_Data!C3)))

Q2: Apply to all 500 rows. Count how many names changed

Q3: =LEN(Raw_Data!C3)-LEN(TRIM(Raw_Data!C3)) → find max spaces

💡 *After writing cleaning formulas → Copy entire column → Paste Special → Values Only. This removes formulas and keeps clean text.*

Step 3 — Handle Blank / Missing Values

CLEANING

Blank cells are dangerous — they break SUM, AVERAGE, VLOOKUP silently. Strategy: Detect → Decide (fill/flag/delete) → Fix. Never delete blindly.

SYNTAX / STEPS

```
=ISBLANK(C3)                → TRUE if truly blank

=OR(ISBLANK(C3),C3="")      → catches both blank types

=COUNTBLANK(C3:C502)        → count blanks in column

=IF(ISBLANK(C3),"Unknown",PROPER(TRIM(C3)))

=IF(ISBLANK(G3),"Not Provided",G3)
```

EXAMPLE DATA

Column	Blank Count	Fill Strategy	Formula
Customer_Name	~15	'Unknown'	=IF(ISBLANK(C3),"Unknown",C3)
Phone	~25	'Not Provided'	=IF(ISBLANK(G3),"Not Provided",G3)
Unit_Price	~8	0	=IF(ISBLANK(Q3),0,Q3)
Product_Name	~12	'Check Required'	=IF(ISBLANK(L3),"Check Required",L3)

FORMULA

```
// Fill blank names:
=IF(ISBLANK(C3),
   "Unknown",
   PROPER(TRIM(C3)))

// Fill blank price:
=IF(ISBLANK(Q3),0,Q3)

// Count all column blanks:
=COUNTBLANK(C3:C502)
```

✏️ EXERCISE

Q1: COUNTBLANK() on each column → build data quality summary table

Q2: CF → New Rule → Is Blank → Orange fill → visual audit

Q3: =IF(OR(ISBLANK(G3),G3=""),"Not Provided",G3)

Step 4 — Fix Date Formats

CLEANING

Mixed date formats (15-Jan-2024, 15/01/2024, 2024-01-15) cause wrong sorting and filtering. Standardize ALL dates to one format before analysis.

SYNTAX / STEPS

```
=TEXT(B3,"DD-MMM-YYYY")          → display format

=DATE(YEAR(B3),MONTH(B3),DAY(B3)) → extract & rebuild

=DAY(B3) | =MONTH(B3) | =YEAR(B3)  → extract parts

=TEXT(B3,"MMMM")                  → month name

=TEXT(B3,"DDDD")                  → day name
```

EXAMPLE DATA

Raw Format	Issue	Fix Method	Clean Output
15/01/2024	DD/MM or MM/DD?	Text to Columns → MDY	15-Jan-2024
2024-01-15	ISO format as text	=DATE(LEFT,MID,RIGHT)	15-Jan-2024
Jan-17-2024	US format	=DATEVALUE(B3)	17-Jan-2024
15-Jan-2024	Correct but mixed	Format Cells → Custom	15-Jan-2024

FORMULA

```
// Extract components:
=DAY(B3)      → 15
=MONTH(B3)    → 1
=YEAR(B3)     → 2024
=TEXT(B3,"MMM") → "Jan"
=TEXT(B3,"DDDD") → "Monday"

// Standardize display:
Format Cells (Ctrl+1)
→ Date → DD-MMM-YYYY
```

✏️ EXERCISE

Q1: Add 3 columns: Order_Day, Order_Month, Order_Year

Q2: =TEXT(B3,"MMMM YYYY") → 'January 2024' for grouping

Q3: Select date column → Ctrl+1 → Custom → DD-MMM-YYYY

Step 5 — Standardize Categories & Text

CLEANING

'Electronics', 'ELECTRONICS', 'Electronicss', 'electronic' — Excel sees these as 4 DIFFERENT values! Standardize before any grouping or Pivot Table.

SYNTAX / STEPS

```
=IF(LOWER(TRIM(L3))="electronics","Electronics",...)
=SWITCH(LOWER(TRIM(I3)),"north","North","south","South"
,...,"Check")
=IF(LOWER(TRIM(U3))="delivered","Delivered",...)
```

Find & Replace (Ctrl+H): 'deliverd' → 'Delivered'

```
=UPPER(A3)  |  =LOWER(A3)  |  =PROPER(A3)
```

EXAMPLE DATA

Raw Value	Issue	Fix Formula	Clean Value
Electronicss	Typo	LOWER+TRIM+IF	Electronics
FASHION	Wrong case	SWITCH formula	Fashion
Soth	Typo	SWITCH formula	Check Required
deliverd	Typo	IF(LOWER...)	Delivered
NO / N / 0	Inconsistent	IF(OR(LOWER...))	No

FORMULA

```
// Category fix:
=IF(LOWER(TRIM(L3))=
   "electronics",
   "Electronics",
   IF(LOWER(TRIM(L3))=
      "fashion","Fashion",
   "Check Required"))

// Region fix with SWITCH:
=SWITCH(LOWER(TRIM(I3)),
  "north","North",
  "south","South",
  "east","East",
  "west","West",
  "Check Required")
```

✏️ EXERCISE

Q1: Write Category clean formula for all 5 valid categories

Q2: Write Region SWITCH formula for North/South/East/West

Q3: COUNTIF for 'Check Required' — what % needs manual review?

Step 6 — Fix Invalid Numbers

CLEANING

Negative quantity, discount > 100%, text in number columns — these give wrong SUM and AVERAGE. Detect with IF, fix systematically.

SYNTAX / STEPS

```
=IF(OR(NOT(ISNUMBER(O3)),O3<1,O3>100),"INVALID","OK")

=IF(R3>80,0,R3)     → cap discount at 80%

=IFERROR(VALUE(SUBSTITUTE(Q3,"₹","")),0) → convert
₹text to number
=ABS(S3-ROUND(O3*Q3*(1-R3/100)*(1+T3/100),0))>1  →
detect wrong totals
=ROUND(O3*Q3*(1-R3/100)*(1+T3/100),0)  → recalculate
Total
```

EXAMPLE DATA

Column	Issue	Detection	Fix
Quantity	Negative / text	=IF(O3<1,"BAD","OK")	=MAX(1,O3)
Discount_%	Over 100%	=IF(R3>100,"BAD","OK")	=MIN(80,MAX(0,R3))
Unit_Price	Text '₹35,000'	=NOT(ISNUMBER(Q3))	=VALUE(SUBSTITUTE...)
Total_Amount	Wrong calc	Compare vs recalc	=Qty*Price*(1-D%)*(1+GST%)

FORMULA

```
// Fix text price:
=VALUE(SUBSTITUTE(
   SUBSTITUTE(Q3," ₹",""),
   ",",""))

// Recalculate Total:
=ROUND(
  Qty * Price
  * (1 - Disc/100)
  * (1 + GST/100),
0)
```

✏️ EXERCISE

Q1: =COUNTIF(O3:O502,"<0") → count negative quantities

Q2: =COUNTIF(R3:R502,">100") → count invalid discounts

Q3: Recalculate Total_Amount for all rows using formula

Data Cleaning — Complete Summary

CLEANING RECAP

After completing all 6 cleaning steps: Raw_Data (500 rows, messy) → Clean_Data (480 rows, analysis-ready). This is the foundation of all further work.

Step	Issue Fixed	Function Used	Rows Fixed
1	Duplicate Records	COUNTIF + Remove Dupes	~20
2	Extra Spaces	TRIM + CLEAN	~45
3	Wrong Casing	PROPER + UPPER + LOWER	~80
4	Blank Values	ISBLANK + IF	~35
5	Date Formats	TEXT + DATEVALUE	~30
6	Invalid Numbers	IF + VALUE + IFERROR	~27

BEFORE vs AFTER

500
Rows (Raw)

480
Rows (Clean)

~237
Issues Found

~237
Issues Fixed

22
Columns Same

0
#REF! Errors

💡 *Complete cleaning checklist in Data_Cleaning_Steps sheet. After all steps → Copy → Paste Special Values to create final Clean_Data sheet.*

SECTION 03

Data Validation

Prevent errors before they happen · Dropdowns · Rules · Formulas

Data Validation — Dropdowns & List Controls

VALIDATION

Data Validation prevents wrong data at entry time — more powerful than fixing after. Dropdown lists ensure ONLY approved values can be entered in a cell.

SYNTAX / STEPS

Data tab → Data Validation → Allow: List → Source

Source option 1: Type values → North,South,East,West

Source option 2: Cell range → =$A$2:$A$5

Source option 3: Named range → =RegionList

Error Alert tab → Title + Message for user-friendly errors

EXAMPLE DATA

Column	Allow	Source	Error Message
Region (I)	List	North,South,East,West	Invalid! Select from dropdown
Category (L)	List	=CategoryList	Invalid Category!
Order_Status (U)	List	Delivered,Shipped...	Select valid status
Return_Flag (V)	List	Yes,No	Select Yes or No only
Payment (T)	List	=PaymentList	Invalid payment mode

FORMULA

```
// Create dropdown from range:
Data → Data Validation
  Allow: List
  Source: =$A$2:$A$5

// Create named range first:
Select A2:A5 → Name Box
→ type "RegionList" → Enter

// Use in validation:
  Source: =RegionList
```

✏️ EXERCISE

Q1: Select I3:I502 → Data Validation → List → North,South,East,West

Q2: Add Error Alert: 'Invalid Region' → 'Select North/South/East/West'

Q3: Try typing 'Soth' → validation should block it with error

Data Validation — Rules & Custom Formulas

VALIDATION

Beyond dropdowns: restrict numbers to valid ranges, dates to past only, text length limits, and custom formula validation for email and phone format.

SYNTAX / STEPS

Whole Number: Allow: Whole Number → Min: 1 → Max: 100

Decimal: Allow: Decimal → Min: 0 → Max: 80 (discount rule)

Date: Allow: Date → End: =TODAY() (no future dates)

Custom Email: =AND(ISNUMBER(FIND("@",E3)),ISNUMBER(FIND(".",E3)))

Custom Phone: =AND(LEN(F3)=10,ISNUMBER(VALUE(F3)))

FORMULA

```
// Email validation formula:
=AND(
  ISNUMBER(FIND("@",E3)),
  ISNUMBER(FIND(".",E3)),
  FIND("@",E3)>1,
  LEN(E3)>6
)

// Phone = exactly 10 digits:
=AND(
  LEN(F3)=10,
  ISNUMBER(VALUE(F3))
)
```

EXAMPLE DATA

Column	Rule Type	Settings	Error Message
Quantity (O)	Whole Number	Min:1, Max:100	Must be 1 to 100
Discount_% (R)	Decimal	Min:0, Max:80	Max discount is 80%
Unit_Price (Q)	Decimal	Greater than 0	Price must be positive
Order_Date (B)	Date	≤ TODAY()	No future dates allowed
Phone (F)	Custom	LEN=10, ISNUMBER	10 digits required

✏️ EXERCISE

Q1: Select O3:O502 → Whole Number → 1 to 100 → test with -5

Q2: Select B3:B502 → Date → ≤ TODAY() → test with future date

Q3: Data → Data Validation → Circle Invalid Data → see all violations

SECTION 04

Data Visualization

5 Chart Types · When to Use · Common Mistakes

Chart Selection Guide — Which Chart for Which Data?

CHART GUIDE

Wrong chart = wrong message. Chart selection is as important as the data itself. Use this guide every time before creating a chart.

📊 Column Chart

Use for: Compare categories

Example: Sales by Region

Avoid: Too many cats (>10)

📉 Bar (Horizontal)

Use for: Ranking or long names

Example: Top 10 Products

Avoid: Time trend data

📈 Line Chart

Use for: Trends over time

Example: Monthly Sales

Avoid: Non-time categories

🥧 Pie Chart

Use for: Parts of a whole (%)

Example: Category Sales Share

Avoid: >6 slices → use Donut

🍩 Donut Chart

Use for: Same as pie + center

Example: Order Status split

Avoid: Too many small slices

🔀 Combo Chart

Use for: Two measures, diff scales

Example: Sales + Growth Rate

Avoid: Single metric only

💡 *Telugu Tip: Wrong chart use cheste data ni misinterpret cheyyadam easy. Time data = Line. Compare = Column. Proportion = Pie. Ranking = Bar.*

Building All 5 Charts — Step by Step

Chart_Data sheet lo pre-built summary tables unnai. Ee tables select chesi charts create cheyyandi. Each chart has specific data and format requirements.

SYNTAX / STEPS

Chart 1: Select Region + Sales → Insert → Clustered Column

Chart 2: Select Month + Sales → Insert → Line with Markers

Chart 3: Select Category + Sales → Insert → 2D Pie

Chart 4: Select Products + Sales → Insert → Bar (Horizontal)

Chart 5: Select Status + Count → Insert → Donut

EXAMPLE DATA

Chart	Data Range	Type	Key Formatting
Sales by Region	A4:B8 (Chart_Data)	Clustered Column	Color each bar differently
Monthly Trend	A13:B25	Line + Markers	Add trendline → Linear
Category Share	A27:B32	2D Pie	% labels, no 3D!
Top 10 Products	A45:B55	Horizontal Bar	Sort largest→smallest
Order Status	A36:B40	Donut	Color: Green=Delivered,Red =Returned

FORMULA

```
// For all charts:
1. Select data from Chart_Data sheet
2. Insert → Charts → choose type
3. Add Chart Title (click title area)
4. Add Data Labels (right-click)
5. Format colors (Chart Design tab)
6. Remove gridlines for clean look
7. Add Axis Titles (Y-axis: ₹ Sales)

// Combo Chart:
Insert → Combo → Custom
Column=Revenue, Line=Growth%
Check: Secondary Axis for Growth
```

✏️ EXERCISE

Q1: Create Sales by Region column chart from Chart_Data!A4:B8

Q2: Right-click → Add Trendline on Monthly line chart

Q3: Move pie chart to its own sheet: Right-click → Move Chart

SECTION 05

Final Dashboard

KPI Cards · Pivot Tables · Charts · Slicers · Timeline

Dashboard Layout — Zones & Structure

DASHBOARD

Oka great dashboard oka question ni 5 seconds lo answer cheyali. Layout plan chesaka start cheyyi — never place elements randomly!

HEADER

Title + Subtitle + Last Updated

KPI CARDS (5)

Total Sales · Orders · Avg Order · Delivered% · Return Rate

MAIN CHART — Column/Line

Revenue by Region or Monthly Trend (largest visual)

SECONDARY — Pie/Table

Category Share or Top 5 Products

SLICERS + TIMELINE

Region · Category · Order Status + Date Timeline

💡 *View → Show → uncheck Gridlines AND Headings for clean professional dashboard look. Do this FIRST before adding anything.*

Pivot Tables — Foundation of the Dashboard

Pivot Table oka second lo 500 rows of data ni summarize chestundi — no formulas needed! This is the most powerful feature in Excel for data analysis.

SYNTAX / STEPS

Step 1: Click anywhere in Clean_Data → Ctrl+T (make it a Table)

Step 2: Insert tab → PivotTable → New Worksheet

Step 3: Drag Region to Rows, Total_Amount to Values

Step 4: Right-click Value → Value Field Settings → Sum / Average

Step 5: Right-click date → Group → Months for monthly trend

FORMULA

```
// 6 Pivots to create:
Pivot 1: Revenue by Region
  Rows: Region
  Values: Sum of Total_Amount

Pivot 2: Monthly Trend
  Rows: Order_Date (Group→Months)
  Values: Sum of Total_Amount

Pivot 3: Category Performance
  Rows: Category
  Values: Sum, Count, Avg

Pivot 4: Order Status
  Rows: Order_Status
  Values: Count of Order_ID
```

EXAMPLE DATA

Pivot	Rows	Values	Use In Dashboard
1 — Region	Region	Sum of Sales, Count of Orders	KPI + Column Chart
2 — Monthly	Date (Months)	Sum of Sales	Line Chart
3 — Category	Category	Sum + Count + Avg	Pie Chart
4 — Status	Order_Status	Count, % Share	Donut Chart
5 — Top Cities	City	Sum of Sales (Top 10)	Bar Chart
6 — Products	Product_Name	Sum of Sales (Top 10)	Bar Chart

✏️ EXERCISE

Q1: Create Pivot 1: Revenue by Region → sort descending

Q2: Group dates by Month in Pivot 2 → see monthly pattern

Q3: Change Value to Average → compare avg order by region

Slicers, Timeline & Report Connections

SLICERS

Slicers are visual filter buttons. Timeline filters by date. Report Connections links ONE slicer to ALL pivot tables — click once, everything updates!

SYNTAX / STEPS

Insert Slicer: PivotTable Analyze → Insert Slicer → select field

Multi-select: Ctrl+Click slicer buttons for multiple values

Clear slicer: Click X button on slicer to remove filter

Timeline: Analyze → Insert Timeline → Order_Date column

Connect ALL: Right-click slicer → Report Connections → tick all pivots

EXAMPLE DATA

Slicer/Filter	Field	Effect	Style Tip
Region Slicer	Region	Filters all 6 pivots + charts	4 columns layout
Category Slicer	Category	Shows selected category only	Match dashboard color
Status Slicer	Order_Status	Filter by Delivered/Returned etc	3 columns layout
Date Timeline	Order_Date	Months/Quarters/Years filter	Timeline style: dark
Combined effect	All connected	ONE click updates entire dashboard!	⭐ Most powerful feature

FORMULA

```
// Step-by-step slicer setup:
1. Click inside Pivot 1
2. PivotTable Analyze tab
   → Insert Slicer
   → Check: Region, Category
3. Right-click Region Slicer
   → Report Connections
   → Check ALL 6 Pivots
4. Same for Category Slicer
5. Insert Timeline → Date
   → Change level: Months

// Now clicking North in
// Region slicer filters
// ALL charts at once!
```

✏️ EXERCISE

Q1: Insert Region slicer → Report Connections → connect to all pivots

Q2: Insert Date Timeline → change to MONTHS view

Q3: Format slicers: Slicer tab → matching color style

KPI Cards & Dashboard Final Polish

DASHBOARD

KPI cards answer 'how are we doing?' instantly. Linked to live formulas — they update automatically when data refreshes. Design is the final step.

SYNTAX / STEPS

Total Sales: =SUM(Clean_Data!S3:S482)

Total Orders: =COUNTA(Clean_Data!A3:A482)

Avg Order: =AVERAGE(Clean_Data!S3:S482)

Delivered%: =COUNTIF(Clean_Data!U3:U482,"Delivered")/COUNTA(Clean_Data!U3:U482)

Return Rate: =COUNTIF(Clean_Data!V3:V482,"Yes")/COUNTA(Clean_Data!V3:V482)

EXAMPLE DATA

KPI Card	Formula	Value	Color
Total Sales	=SUM(Total_Amount)	₹54,70,000	22D3EE
Total Orders	=COUNTA(Order_ID)	480	34D399
Avg Order Value	=AVERAGE(Total_Amount)	₹11,396	3B82F6
Delivered %	=COUNTIF(..,"Delivered")/COUNTA	76.0%	34D399
Return Rate	=COUNTIF(Return_Flag,"Yes")/COUNTA	6.0%	FBBF24

FORMULA

```
// KPI Card design steps:
1. Insert → Shape → Rectangle
   (1.8" × 1.2" size)
2. Fill: Dark (#1E293B)
3. Add text box inside:
   =SUM(Clean_Data!S3:S482)
   Font: 26pt, Cyan, Bold
4. Add label text below:
   "Total Sales" — 9pt, muted
5. Add delta arrow: ▲+12.3%
   Color: Green if positive

// Design polish:
View → Gridlines: OFF
View → Headings: OFF
Background: Navy (#1B2A4A)
Font: Consistent Calibri
```

✏️ EXERCISE

Q1: Create 5 KPI cards using shapes + linked formulas

Q2: Hide gridlines + headings (View → Show → uncheck)

Q3: Review → Protect Sheet → allow Select Locked Cells only

Dashboard Do's & Don'ts — Design Rules

DESIGN RULES

Professional dashboards follow design rules. Breaking these creates confusion and looks unprofessional — even if data is correct.

✅ DO

- ✓ Consistent 2-3 colour palette (navy+teal+white)
- ✓ Lead with most important KPI at top-left
- ✓ Label every axis with units (₹, %, count)
- ✓ Add slicers for interactivity
- ✓ Leave breathing space — don't fill every inch
- ✓ Test with a real user before sharing
- ✓ Hide gridlines + headings for clean look
- ✓ Protect sheet to prevent accidental edits

❌ DON'T

- ✗ Never use 3D charts — they distort data
- ✗ Never use more than 5 colours at once
- ✗ Never pie chart with >6 slices
- ✗ Never show #REF! or #N/A errors
- ✗ Never hardcode KPI values
- ✗ Never skip chart titles and axis labels
- ✗ Never use dark text on dark backgrounds
- ✗ Never share without hiding raw data sheets

💡 *One dashboard = one question. Ask: 'What should the viewer understand in 5 seconds?' Design everything around THAT one answer.*

SECTION 06

Interview Questions & Assignment

Prepare for job interviews + Practice assignment

Top Interview Questions — Excel Data Analytics

INTERVIEW PREP

Ee questions job interviews lo frequently ask chesthaaru — especially data analyst, MIS analyst, business analyst roles lo. Telugu lo nerchukunnaru — English lo answer cheyyadam practice cheyyi!

Fresher Level

1. What is TRIM function? When do you use it?
2. How do you find and remove duplicates in Excel?
3. Explain ISBLANK vs LEN=0 difference
4. What is data validation? Give a real example
5. Difference between SUMIF and SUMIFS?

Intermediate Level

1. How do you standardize 1000 rows of inconsistent city names?
2. What is the difference between PivotChart and regular chart?
3. How do you connect one slicer to multiple pivot tables?
4. If Order_Date is stored as text, how do you convert it?
5. Write a formula to validate email format in Excel

Advanced Level

1. How would you build a dashboard that auto-updates with new data?
2. Explain Report Connections in the context of slicers
3. How would you use Power Query instead of TRIM+PROPER formulas?
4. What is a named range? How does it help dynamic charts?
5. Design a data cleaning workflow for 10,000 row HR dataset

💡 *Telugu Tip: Har question ki answer ready cheyyi. Excel lo demonstrate cheyyadam chala important — theory alone sufficient kadu!*

Mini Project Assignment — 100 Marks

ASSIGNMENT

Assignment Dataset: 100-row Student Performance data (messy). Clean, validate, visualize and build a mini dashboard. Screenshot → Comment in YouTube!

MARKS	TASK	DETAILS
10	TRIM + PROPER on Student Names	Add clean name column
10	Remove Duplicate Student_IDs	Use COUNTIF + Remove Dupes
10	Handle Blank Names	Fill with 'Unknown Student'
10	Fix Invalid CGPA (>10 or <0)	Flag as 'Check Required'
10	Standardize Gender (M→Male etc.)	Use IF/SWITCH formula
10	Data Validation on Placement_Status	Dropdown: Placed/Not Placed
15	Chart: Avg CGPA by Branch	Clustered Column Chart
15	Chart: Placement Rate by College	Pie/Donut Chart
10	Mini KPI Dashboard	3 KPI cards + 2 charts

📤 HOW TO SUBMIT

1. Download Assignment sheet from Excel file
2. Complete all 9 tasks
3. Take screenshot of dashboard
4. Comment screenshot in YouTube video
5. Best submissions = featured next video!

⏰ **Deadline: 7 days from video publish**

📁 **File link in description**

❓ **Doubts? Comment below!**

PROJECT COMPLETE!

You can now clean, validate

& present any dataset!

🔍 **Raw Dataset**	500-row messy e-commerce data — 22 cols, all real issues
✅ **Validation**	Dropdowns, number rules, date rules, email/phone formulas
📈 **Pivot Tables**	6 pivots summarizing 480 clean rows instantly
📊 **Dashboard**	KPI cards + charts + slicers — interactive, auto-updates
🧹 **Data Cleaning**	6 steps: TRIM+PROPER+ISBLANK+Dates+Categories+Numbers
📊 **Visualization**	5 charts: Column+Line+Pie+Bar+Donut — chart selection guide
🎛️ **Slicers**	Region+Category+Status+Timeline — all connected!
🎯 **Assignment**	100-mark student project to prove your skills

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